



Lect 01

Introduce active filter circuits

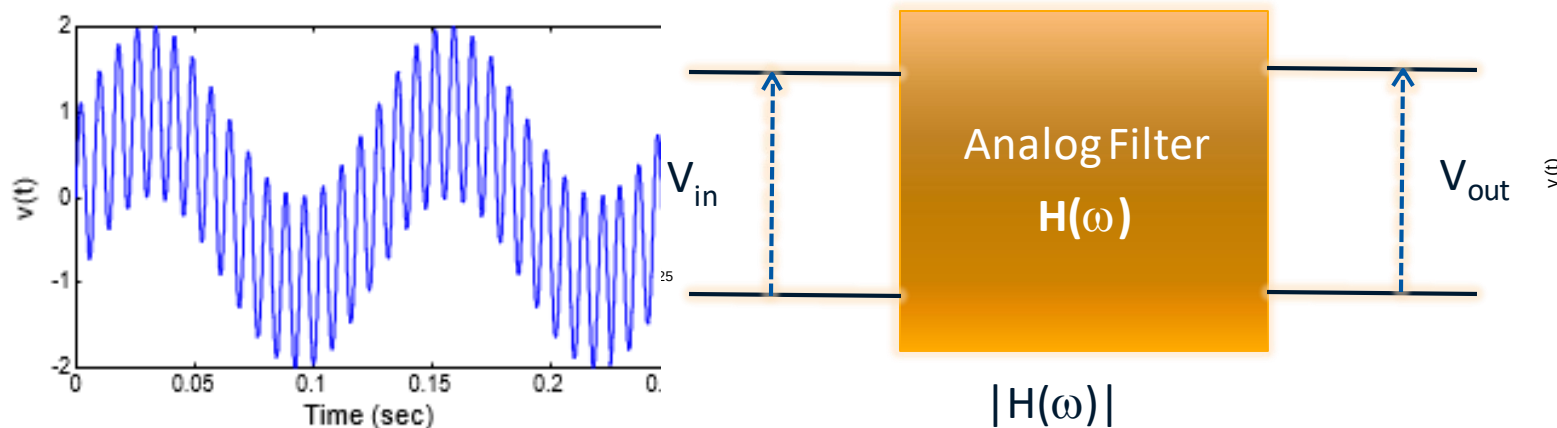
Instructor /Noor .H .Jabber





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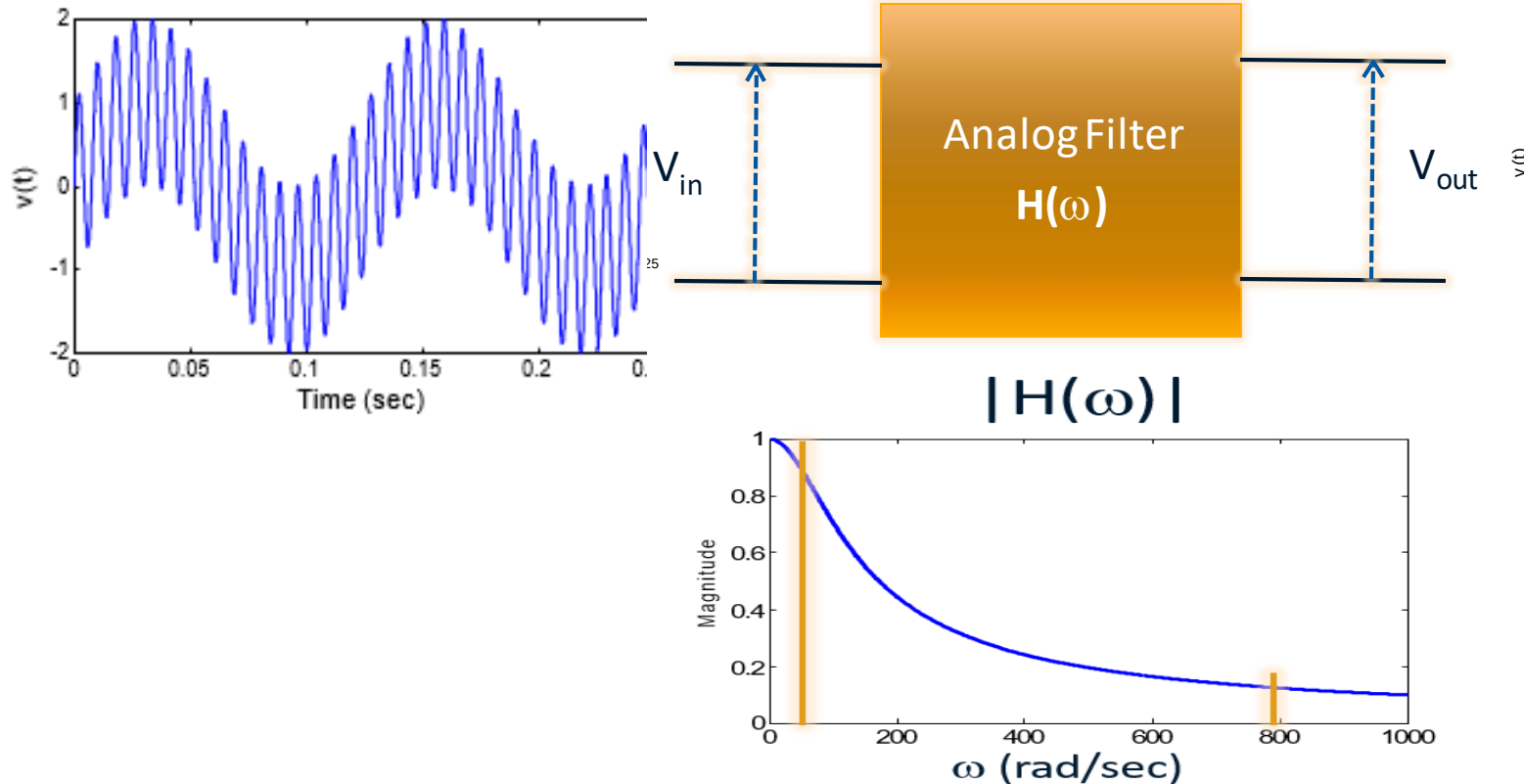
Analog Filters





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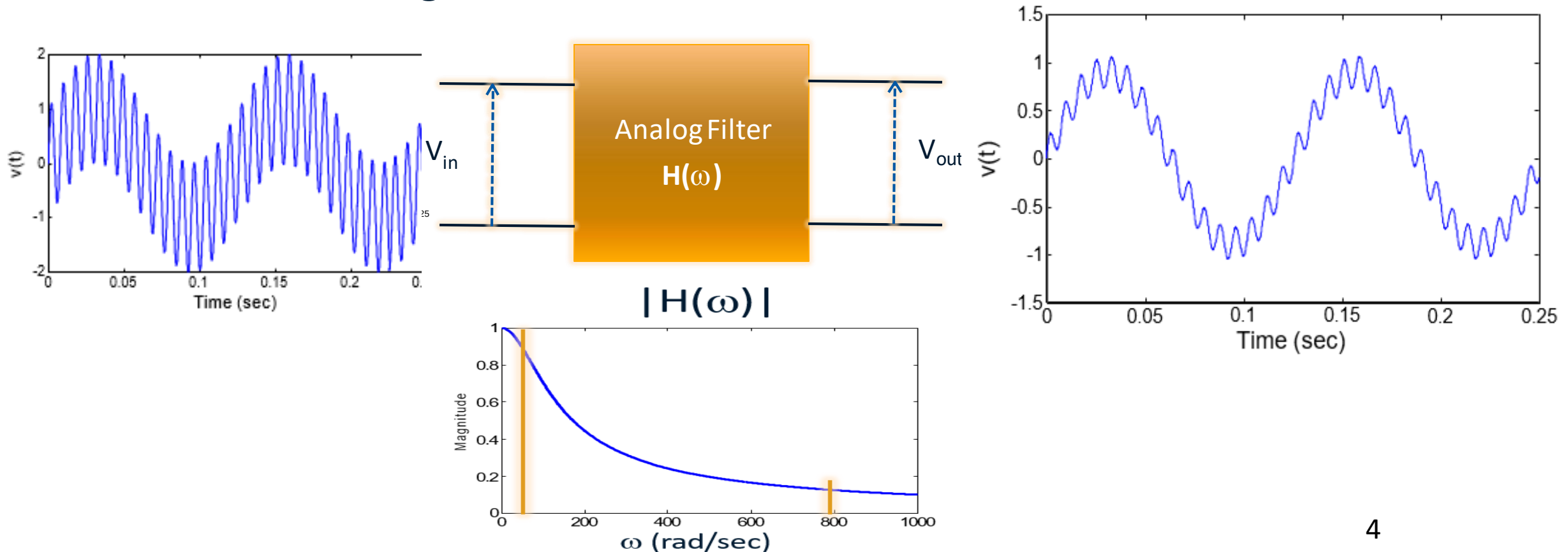
Analog Filters





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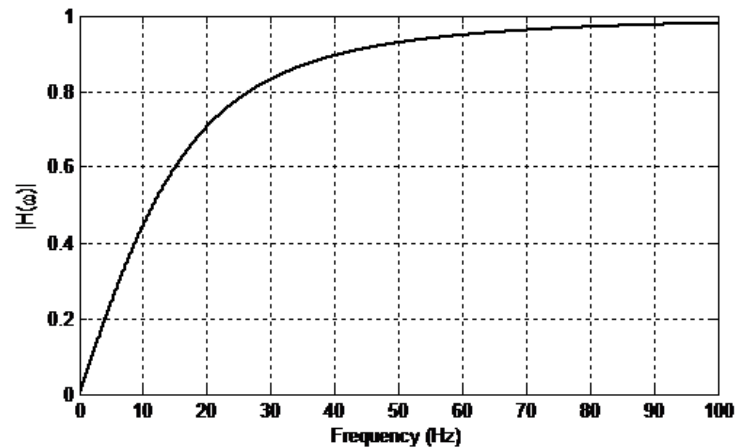
Analog Filters



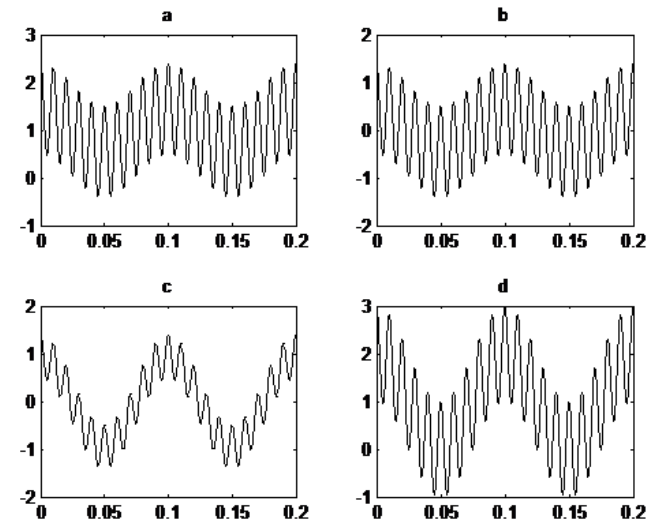


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Quiz



$$V_{in} = 1 + \cos(10(2\pi t)) + \cos(100(2\pi t))$$



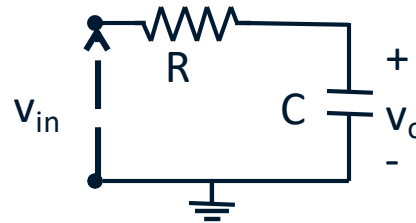
$$V_{out} = 0.45\cos(10(2\pi t) + \theta_1) + 0.97\cos(100(2\pi t) + \theta_2)$$



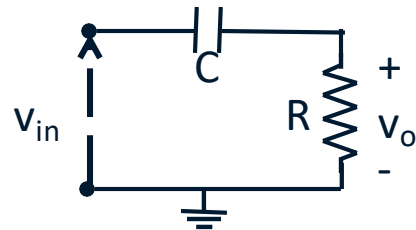
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Summary of RC and RLC (Passive) Filters

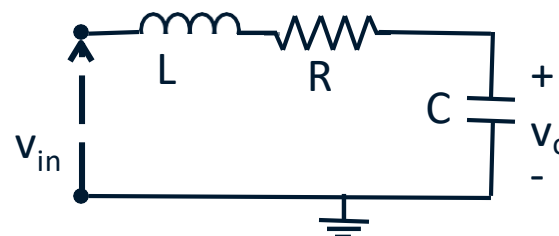
RC Lowpass:



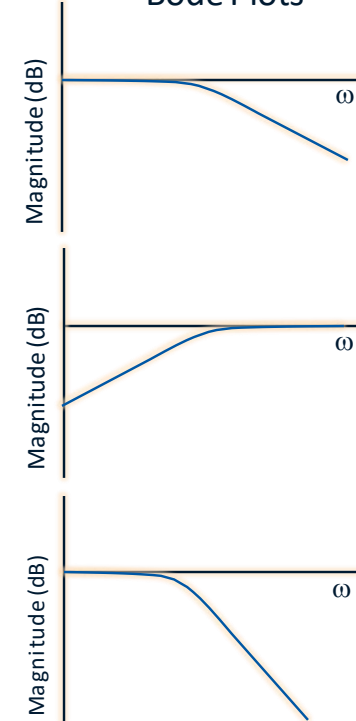
RC Highpass:



RLC Lowpass:



Bode Plots

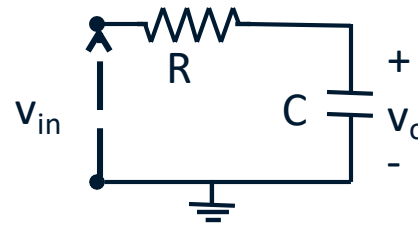




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Limitations of RLC Passive Filters

- Depletes power
- No isolation





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Active Filters

Active – has its own power supply

- Most common active filters are made from op amps
- Provide isolation





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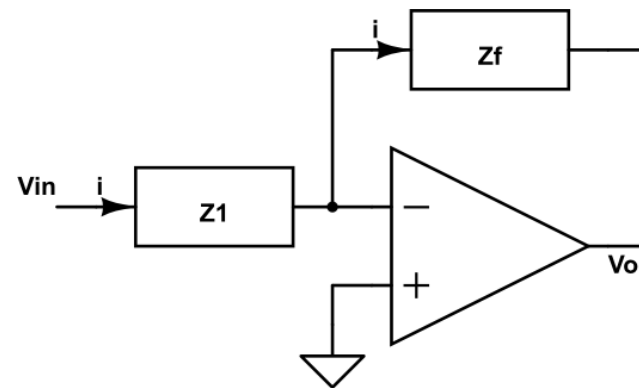
Summary

- ◉ An **analog filter** is a circuit that has a specific shaped frequency response
- ◉ A **active filter** is made of op amps and has its own power supply.
Advantages over RLC passive filters:
 - Provides isolation (cascade filters)
 - Boosts the power
 - Can provide sharper roll-off



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Impedance Gain



$$V_o = \frac{-Z_F}{Z_{in}} V_{in}$$

Derivation: $V_{in} = iZ_1$

$$V_o = -iZ_f = -(Z_f/Z_1)V_{in}$$



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Question

What are the main advantages of an active filter over a passive filter?

- ☒ Provides isolation
- ☒ Can have a sharper roll-off
- ☒ Boosts the power
- ☐ Uses less power



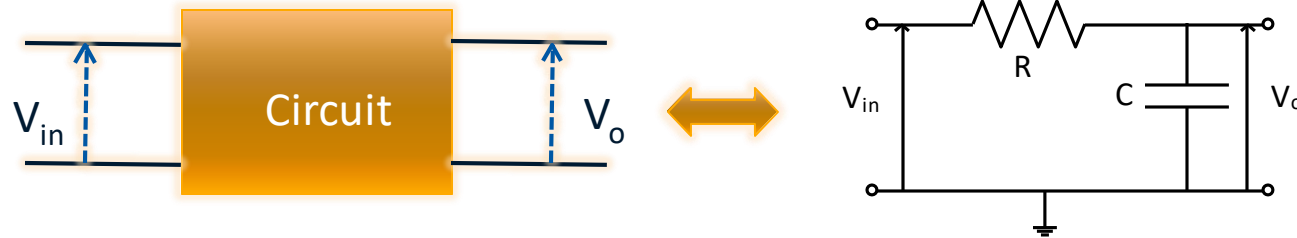
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1. Which of the following are true about active filters as compared to passive filters. Choose all that apply.

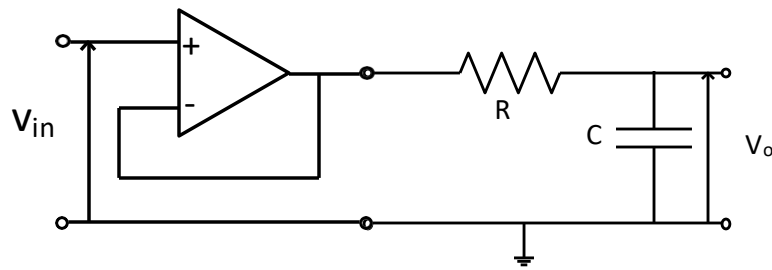
- ☒ Active filters have their own power source.
- ☐ Active filters cannot realize the same transfer functions as passive filters.
- ☒ Active filters can have a DC gain greater than unity, unlike passive filters.
- ☐ Active filters can be cascaded without affecting their transfer functions, unlike passive filters.
- ☐ Active filters behave linearly for all input values.

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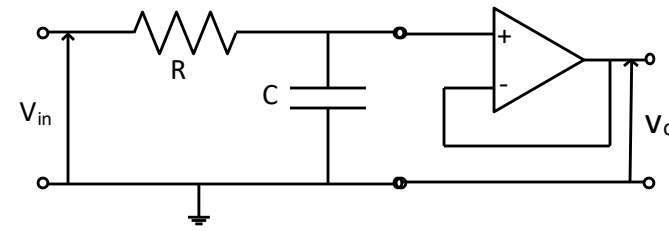
From Passive to Active Lowpass Filters



Isolation at the input:

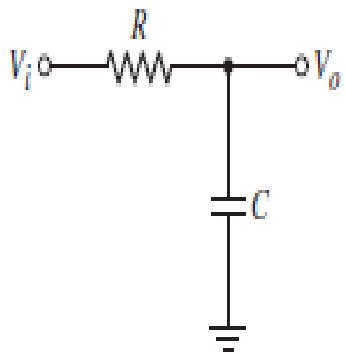


Isolation in the output:

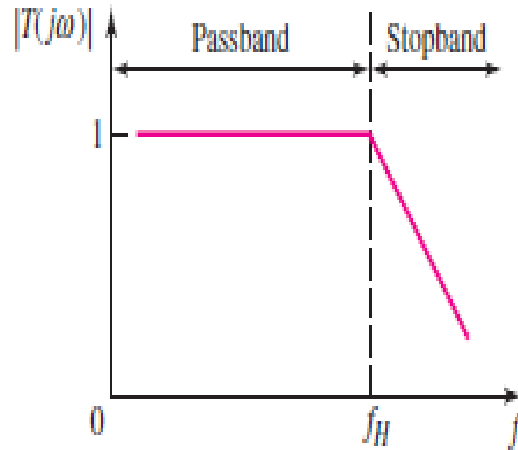




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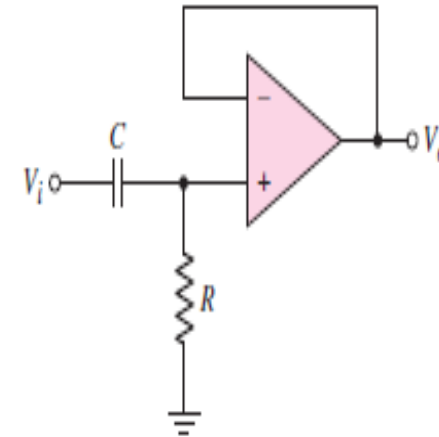


(a)

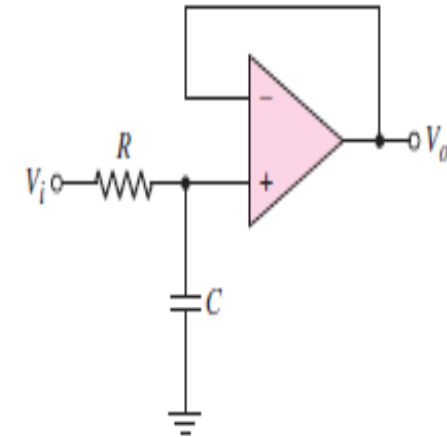


(b)

(a) Simple low-pass filter and (b) Bode plot of transfer function magnitude



(a)



(b)

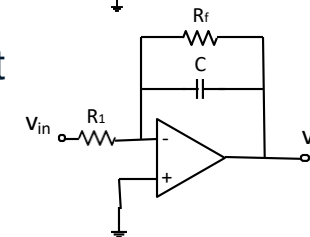
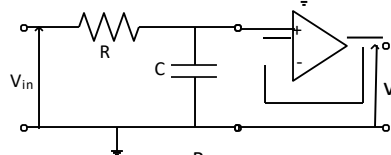
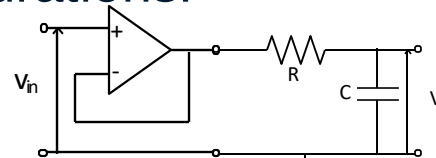
a) High-pass filter with voltage follower and (b) low-pass filter with voltage follower



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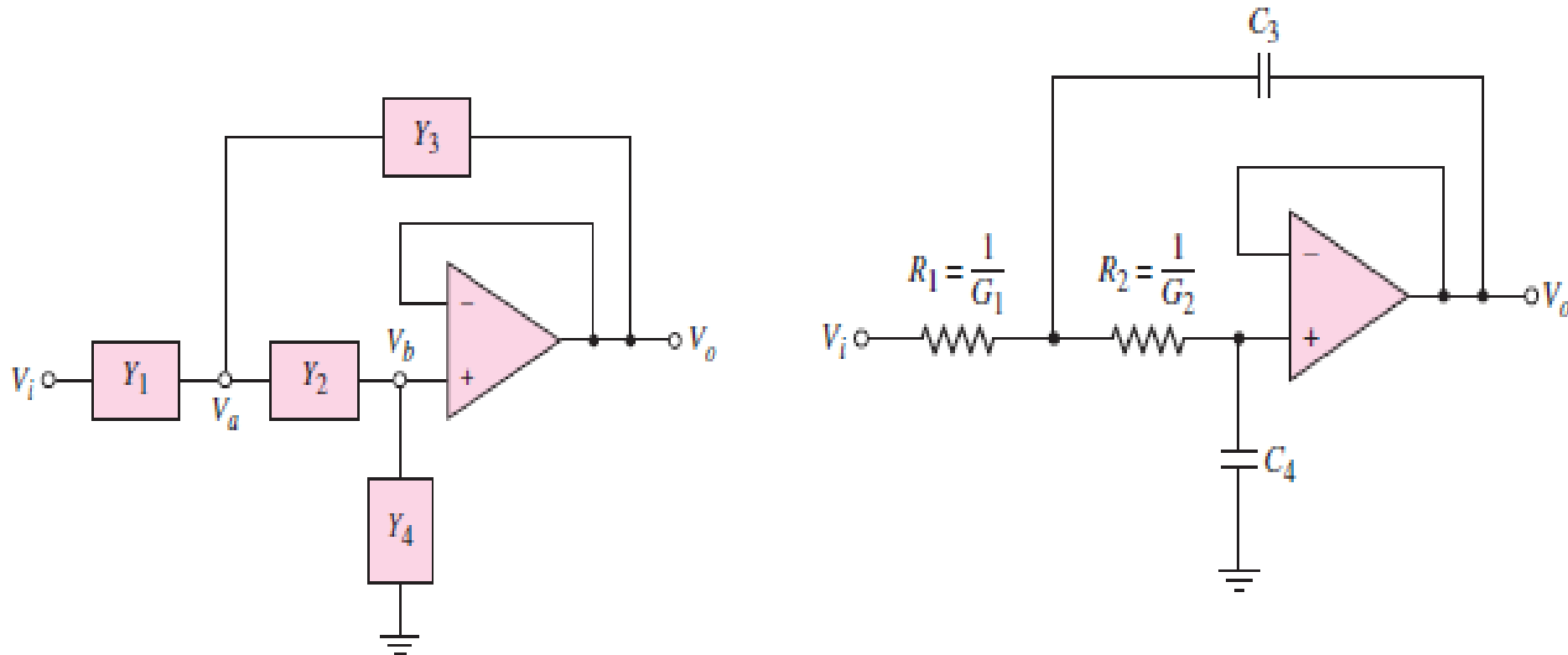
Summary

- ◉ A **lowpass filter** passes low frequency signals and attenuates high frequency signals
- ◉ Three first-order lowpass configurations:
 - Noninverting, isolation at the input
 - Noninverting, isolation at the output
 - Inverting, isolation at input and output





General Two-Pole Active Filter

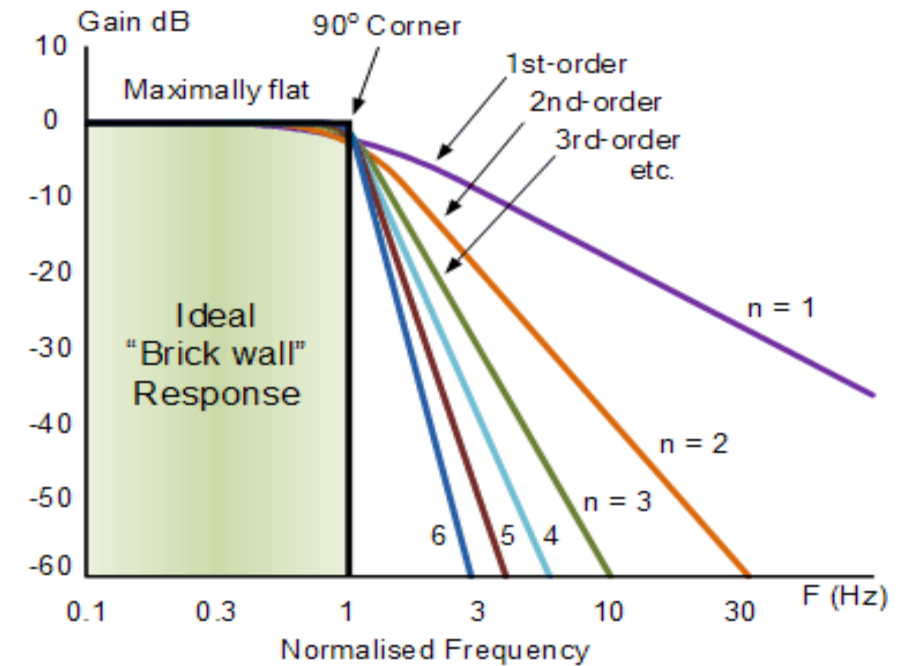
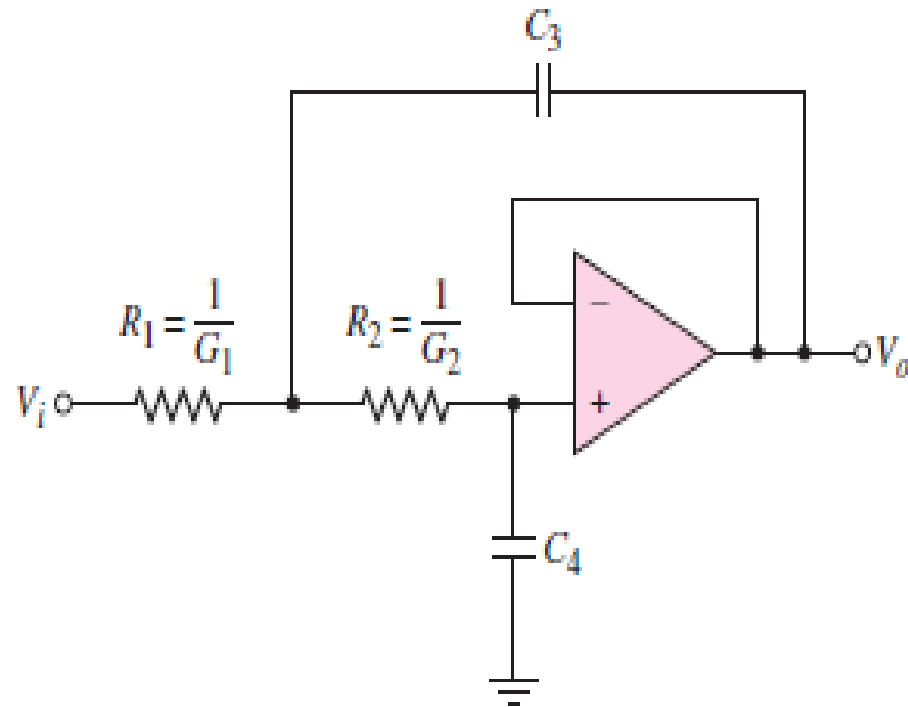




Analog Electronics Lab



General Two-Pole Active Filter





Analog Electronics Lab



Next lesson

Objective: Design a two-pole low-pass Butterworth filter for an audio amplifier application.

Specifications: The circuit with the configuration shown in Figure 15.7(a) is to be designed such that the bandwidth is 20 kHz.

Choices: An ideal op-amp is available and standard-valued resistors and capacitors must be used.

